

Enable Multi-Factor Authentication

[Edition 09]

[Last Update 240224]

Scenario: Imagine a growing tech startup, "InnovateXYZ," that has been rapidly expanding its product offerings. The startup utilizes AWS to host its web applications, manage its databases, and store sensitive customer data. As the company grows, so does the number of employees who need access to these AWS resources. So, to enhance security, your company decides to implement Multi-Factor Authentication (MFA) for all AWS accounts within the company. MFA requires users to provide two or more verification factors to gain access to AWS resources, making unauthorized access significantly more difficult.

Description: The AWS Multi-Factor Authentication (MFA), a security measure that adds an extra layer of protection to your AWS account by requiring both your password and a code from your MFA device. The included guide covers enabling MFA on the root account, signing into the AWS Console with MFA, and deactivating MFA. No additional fees are charged by AWS for using MFA.

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1 INTRODUCTION

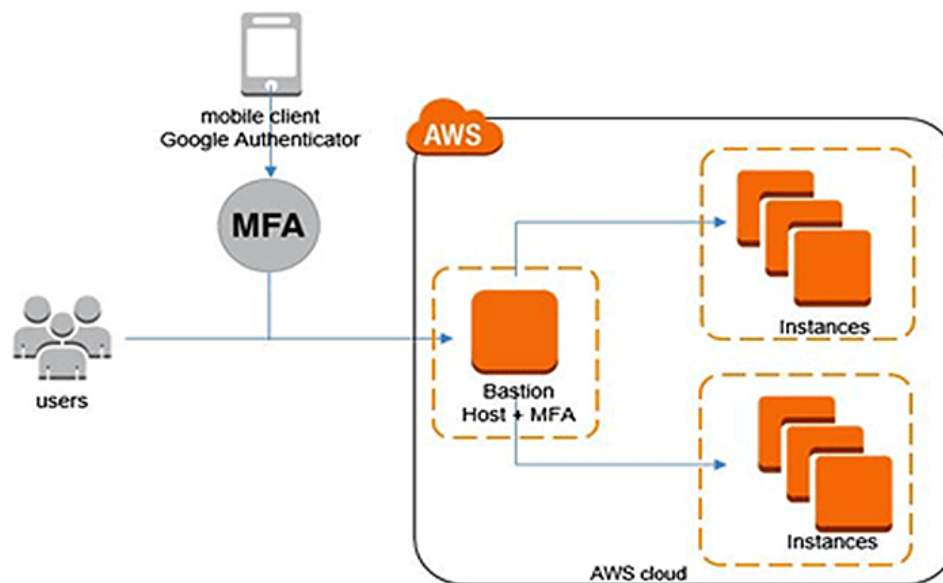
AWS Multi-Factor Authentication (MFA) is a simple best practice that adds an extra layer of protection on top of your username and password. With MFA enabled, when a user signs in to an AWS Management Console, they will be prompted for their user name and password (the first factor-what they know), as well as for an authentication code from their AWS MFA device (the second factor-what they have). Taken together, these multiple factors provide increased security for your AWS account settings and resources.

You can enable MFA for your AWS account and for individual IAM users you have created under your account. MFA can also be used to control access to AWS service APIs.

After you've obtained a supported hardware or virtual MFA device, AWS does not charge any additional fees for using MFA.

This activity guide covers steps:

- Enable MFA on the Root account.
- Accessing AWS Console using MFA
- Deactivating of MFA



2 OVERVIEW & KEY CONCEPTS

In this section, we'll cover key concepts used in this guide, including an overview of what's covered in this guide.

2.1 What is MFA?

AWS multi-factor authentication (MFA) is a recommended practice for AWS Identity and Access Management (IAM) that needs an additional authentication factor in addition to the user name and password sign-in credentials. MFA may be enabled at the AWS account level as well as for root and IAM users in your account. AWS is increasing the eligibility requirements for its free MFA security key program. Check your eligibility and request your free MFA key.

When a user signs in to the AWS Management Console with MFA enabled, they are requested for their user name and password, which they know, as well as an authentication code from their MFA device, which they have. These characteristics, when combined, increase AWS security.

MFA types represent the different mechanisms that users will be able to register and provide as secondary means of verifying their identity when challenged. All MFA types are supported for both browser-based console access as well as using the AWS CLI v2 with IAM Identity Center.

IAM Identity Center MFA provides support to enable one or both of the following types of client-side authentication types. A user can have up to eight MFA devices registered to one account.

- **Authenticator apps**
- **Security keys and built-in authenticators**

Documentation: [Learn more about MFA](#)

Blog: <https://k21academy.com/amazon-web-services/aws-multi-factor-authentication-mfa/>

3 DOCUMENTATION LINKS

1. Multi-Factor Authentication
<https://docs.aws.amazon.com/singlesignon/latest/userguide/enable-mfa.html>
2. Enabling MFA devices for users in AWS
https://docs.aws.amazon.com/IAM/latest/UserGuide/id_credentials_mfa_enable.html

4 PREREQUISITE

1. An AWS Account (Free or Paid)

K21Academy

5 Multi Factor Authentication on Root Account

Using **Multi-Factor Authentication (MFA)** on the root account is an important security measure for protecting AWS (Amazon Web Services) **resources**. The root account is the most privileged account in an AWS account, and has full access to all resources, billing information, and user management functions.

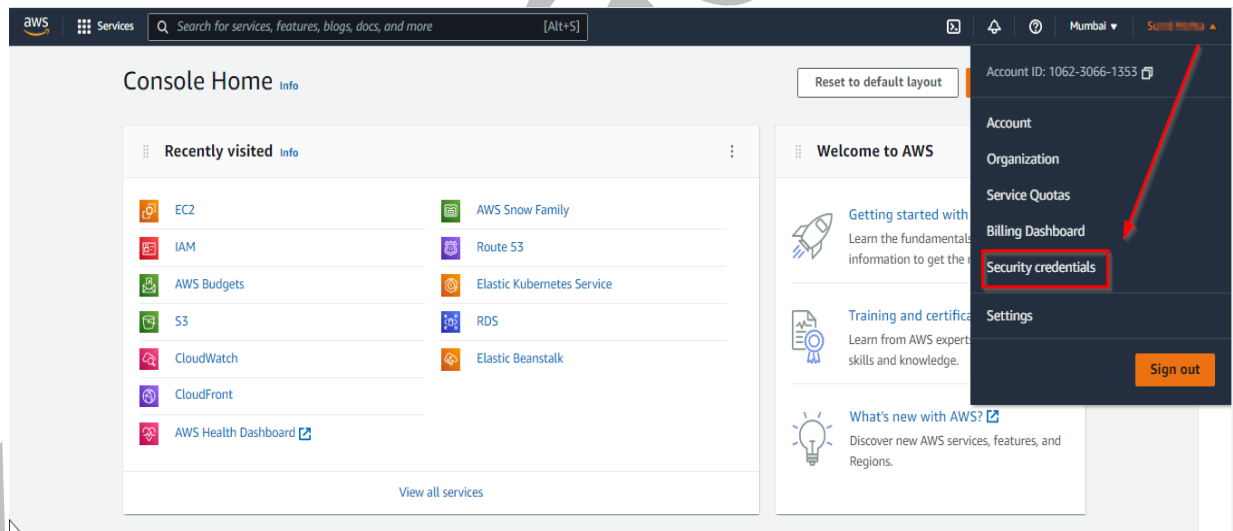
Enabling MFA on the root account adds **an extra layer of security** to the authentication process, making it more difficult for unauthorized individuals to access the account

In this section we will be covering steps to Enable and access your AWS account using Multi factor Authentication and also for the deactivation of MFA from AWS account.

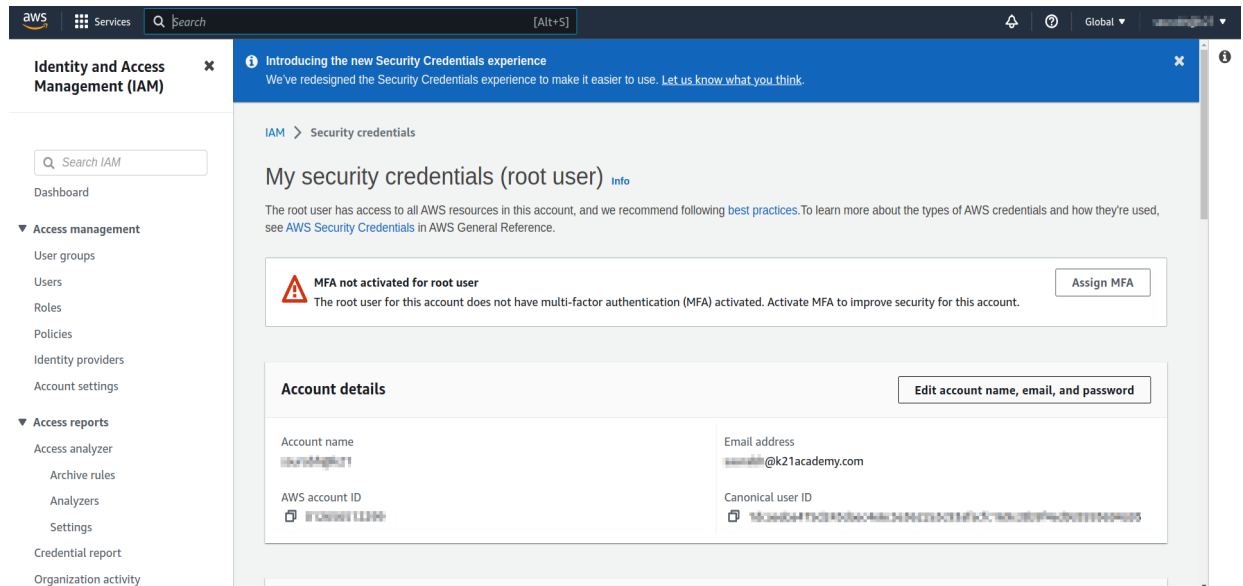
5.1 Enable Multi Factor Authentication on Root Account

Note: In this section we will cover how to enable an MFA on your AWS account

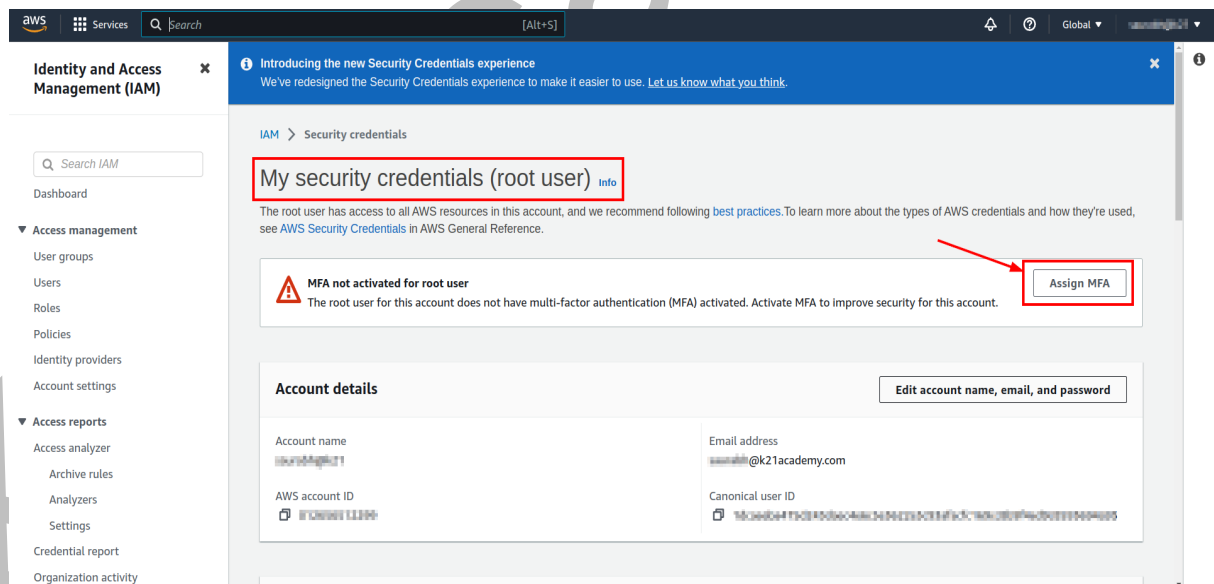
1. On the right side of the navigation bar, choose your account name, and choose **My Security Credentials**.



2. Now you shall see this screen.



3. Now Click on Assign MFA.



4. **Specify MFA device name**, select MFA device as **Authenticator app** and click on **Next**.

The screenshot shows a two-step wizard. The first step, 'Specify MFA device name', has a text input field containing 'k21-mfa' and a 'Next' button. The second step, 'Select MFA device', shows three options: 'Authenticator app' (selected), 'Security Key', and 'Hardware TOTP token'. A red arrow points to the 'Next' button at the bottom right of the second step.

Specify MFA device name

Device name
Enter a meaningful name to identify this device.

k21-mfa

Maximum 128 characters. Use alphanumeric and '+ = , . @ - _ ' characters.

Select MFA device [Info](#)

Select an MFA device to use, in addition to your username and password, whenever you need to authenticate.

☒  **Authenticator app**
Authenticate using a code generated by an app installed on your mobile device or computer.

☐  **Security Key**
Authenticate using a code generated by touching a YubiKey or other supported FIDO security key.

☐  **Hardware TOTP token**
Authenticate using a code displayed on a hardware Time-based one-time password (TOTP) token.

Cancel **Next**

5. Now Install Google Authenticator on your phone.

Android:

https://play.google.com/store/apps/details?id=com.google.android.apps.authenticator2&hl=en_IN

IOS:

<https://apps.apple.com/us/app/google-authenticator/id388497605>

6. Now Click on Show QR Code and open the Google Authenticator app in your phone.

Set up device

Set up your authenticator app

A virtual MFA device is an application running on your device that you can configure by scanning a QR code.

1

Install a compatible application such as Google Authenticator, Duo Mobile, or Authy app on your mobile device or computer.

[See a list of compatible applications](#) 

2



Open your authenticator app, chose **Show QR code** on this page, then use the app to scan the code. Alternatively, you can type a secret key.

[Show secret key](#)

Fill in two consecutive codes from your MFA device.

3

MFA code 1

MFA code 2

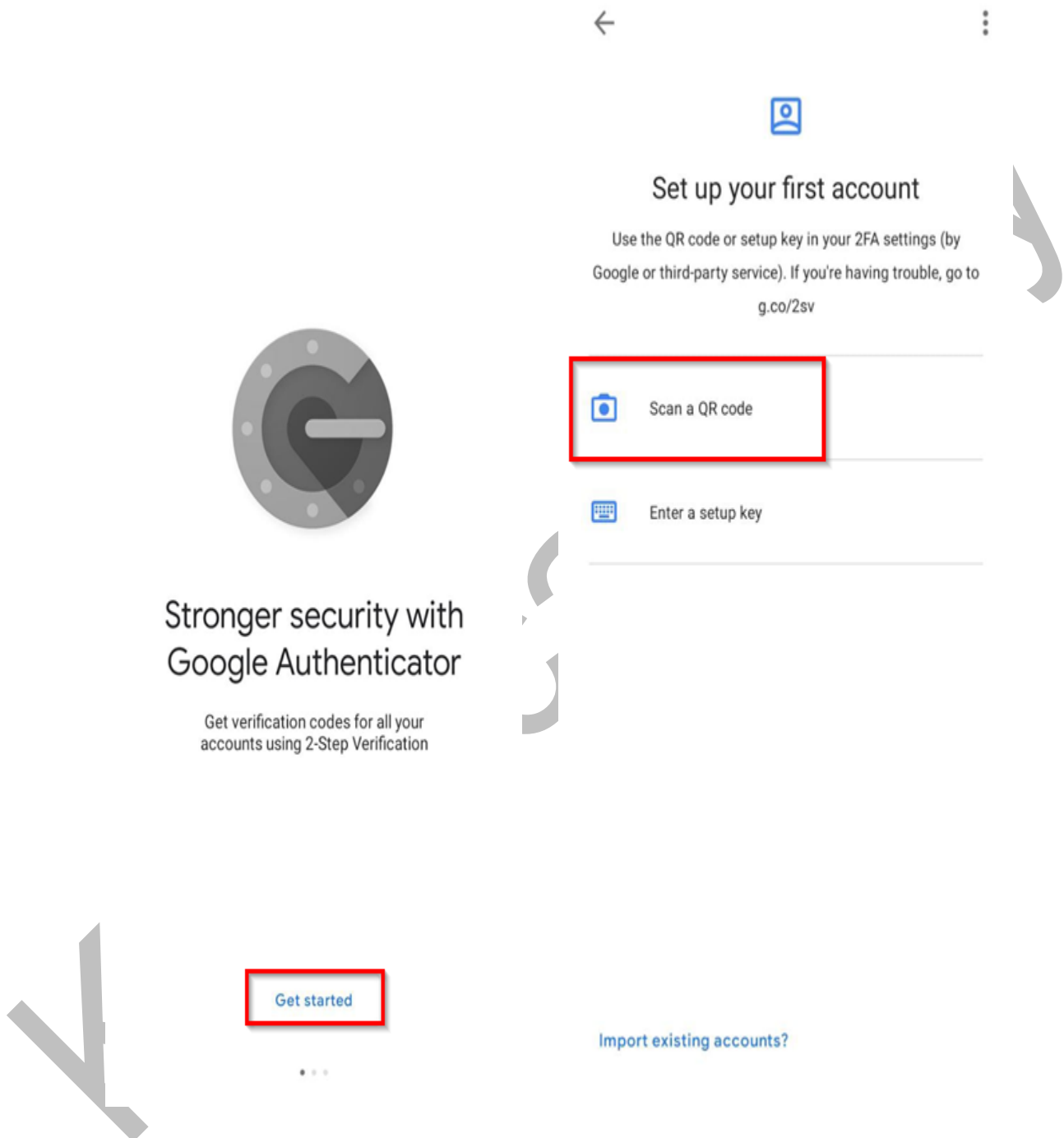
Cancel

Previous

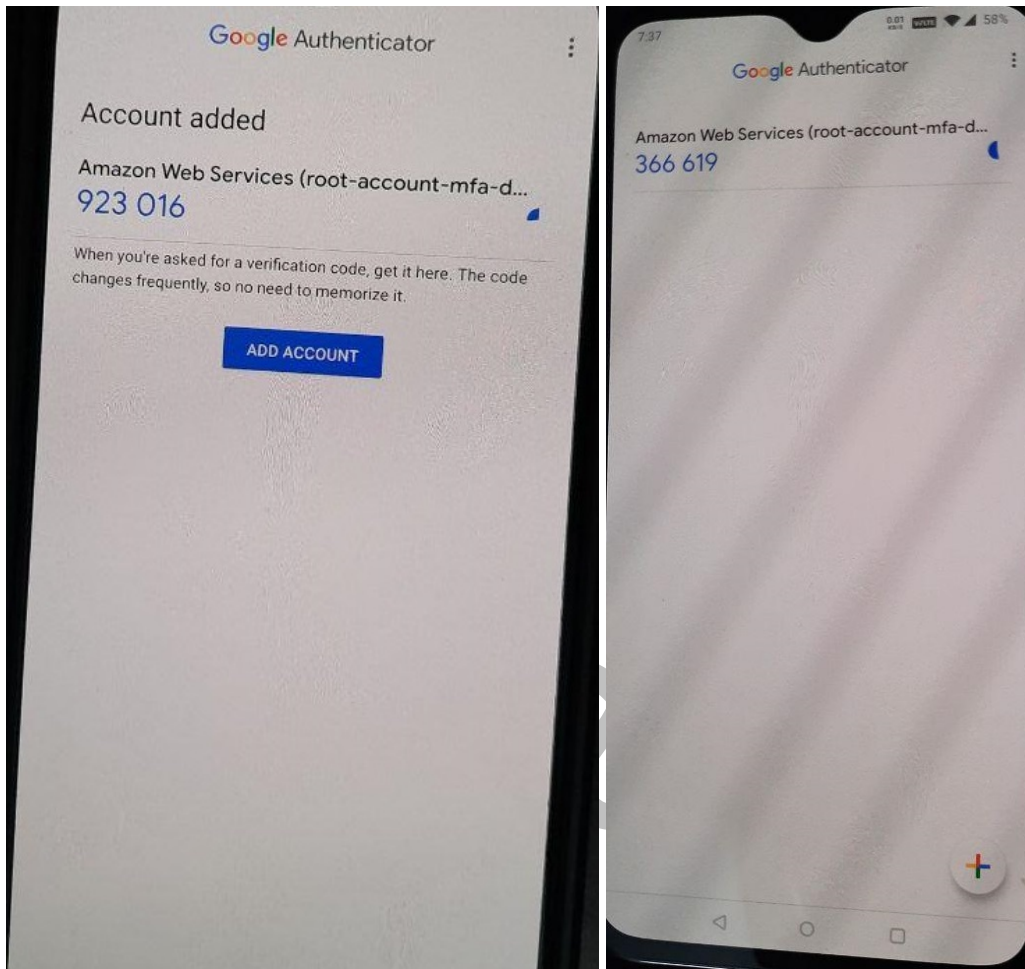
Add MFA

Note: Take a screenshot of the code so that in future if you lose your phone you can use it to re-enable MFA.

7. Now open the Google Authenticator App Click on Get started and Scan the QR code.



8. Now Enter the code from your Phone into MFA code 1 and MFA code 2.



9. After adding MFA code click on **Assign MFA**.

Set up device

Set up your authenticator app

A virtual MFA device is an application running on your device that you can configure by scanning a QR code.

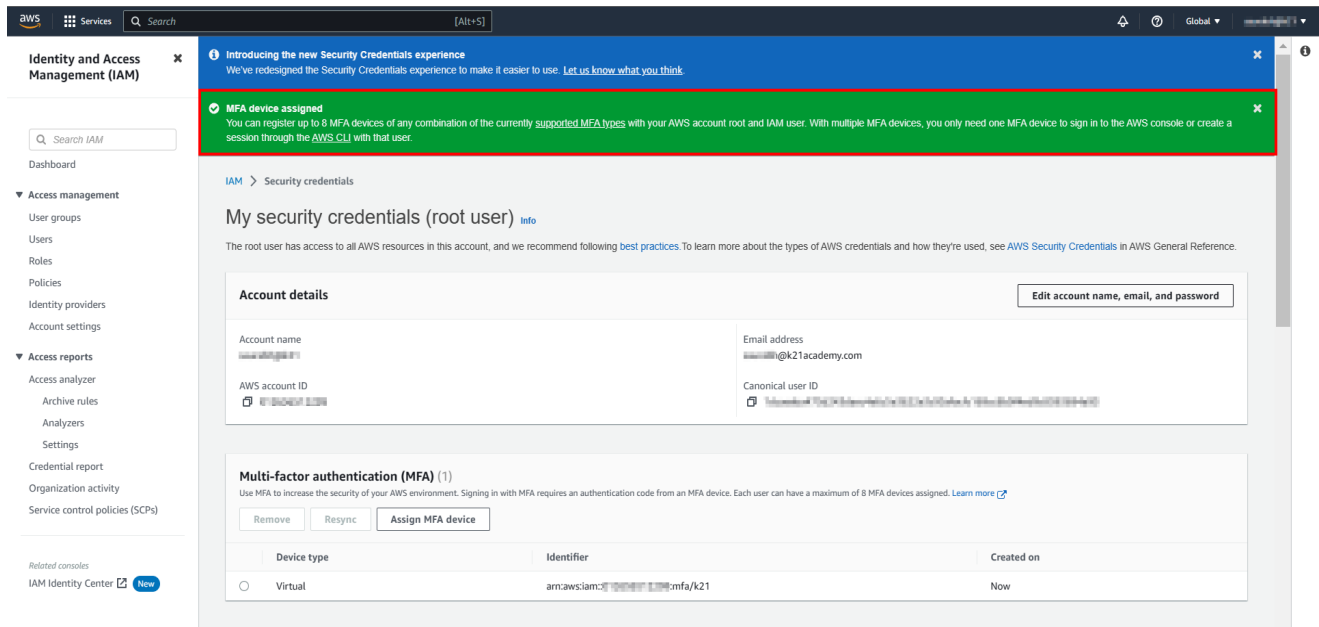
- 1 Install a compatible application such as Google Authenticator, Duo Mobile, or Authy app on your mobile device or computer.
[See a list of compatible applications](#)
- 2 
Open your authenticator app, chose **Show QR code** on this page, then use the app to scan the code.
Alternatively, you can type a secret key. [Show secret key](#)
- 3 Fill in two consecutive codes from your MFA device.
MFA code 1

MFA code 2

Cancel Previous **Add MFA**

Note: Your MFA codes can be different than those shown in above screenshots. If you face any errors at this step, please check the solution in **Troubleshooting Section**

10. You will get a success message.



Introducing the new Security Credentials experience
We've redesigned the Security Credentials experience to make it easier to use. [Let us know what you think](#)

MFA device assigned
You can register up to 8 MFA devices of any combination of the currently supported MFA types with your AWS account root and IAM user. With multiple MFA devices, you only need one MFA device to sign in to the AWS console or create a session through the AWS CLI with that user.

My security credentials (root user) [Info](#)

The root user has access to all AWS resources in this account, and we recommend following [best practices](#). To learn more about the types of AWS credentials and how they're used, see [AWS Security Credentials](#) in AWS General Reference.

Account details [Edit account name, email, and password](#)

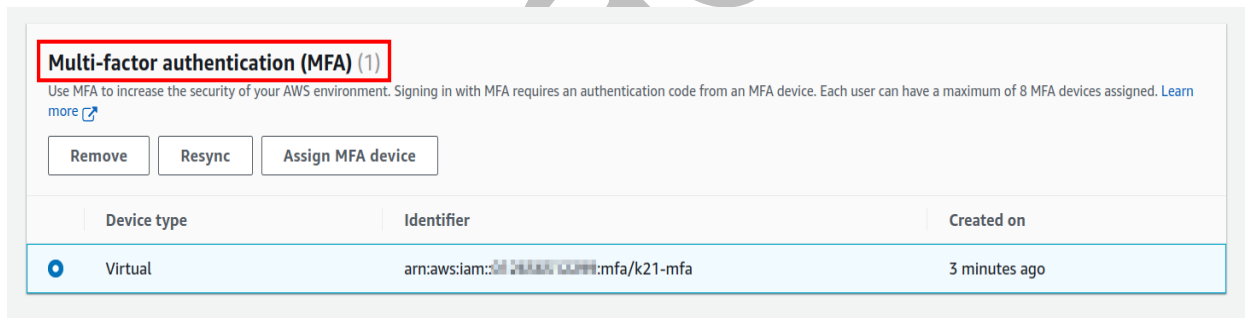
Account name [redacted]	Email address [redacted]@k21academy.com
AWS account ID [redacted]	Canonical user ID [redacted]

Multi-factor authentication (MFA) (1)
Use MFA to increase the security of your AWS environment. Signing in with MFA requires an authentication code from an MFA device. Each user can have a maximum of 8 MFA devices assigned. [Learn more](#)

[Remove](#) [Resync](#) [Assign MFA device](#)

Device type	Identifier	Created on
☐ Virtual	arn:aws:iam::[redacted]:mfa/k21	Now

11. Now you will See that the device has been added for MFA.



Multi-factor authentication (MFA) (1)
Use MFA to increase the security of your AWS environment. Signing in with MFA requires an authentication code from an MFA device. Each user can have a maximum of 8 MFA devices assigned. [Learn more](#)

[Remove](#) [Resync](#) [Assign MFA device](#)

Device type	Identifier	Created on
☒ Virtual	arn:aws:iam::[redacted]:mfa/k21-mfa	3 minutes ago

Note: Now you have successfully Activated MFA on your root account setting.

5.2 Access AWS Console Using MFA

Note: In this section we are going to Access our AWS account in which we enabled the MFA.

1. Open your AWS console login page and click on **Root User** then enter you email address associated with AWS Account



Sign in

☒ **Root user**

Account owner that performs tasks requiring unrestricted access. [Learn more](#)

☐ **IAM user**

User within an account that performs daily tasks. [Learn more](#)

Root user email address

Next

By continuing, you agree to the [AWS Customer Agreement](#) or other agreement for AWS services, and the [Privacy Notice](#). This site uses essential cookies. See our [Cookie Notice](#) for more information.

— New to AWS? —

Create a new AWS account



2. Enter your password corresponding to Email address

aws

Root user sign in

Email: sumtimehta438@gmail.com

Password [Forgot password?](#)

Sign in

[Sign in to a different account](#)

[Create a new AWS account](#)



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English

3. Use your **Google Authenticator** Application on mobile and enter MFA code in AWS Console

aws

Multi-factor authentication

Your account is secured using multi-factor authentication (MFA). To finish signing in, turn on or view your MFA device and type the authentication code below.

Email address: sumtimehta438@gmail.com

MFA code

889323

Submit

[Troubleshoot MFA](#)

[Cancel](#)



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English

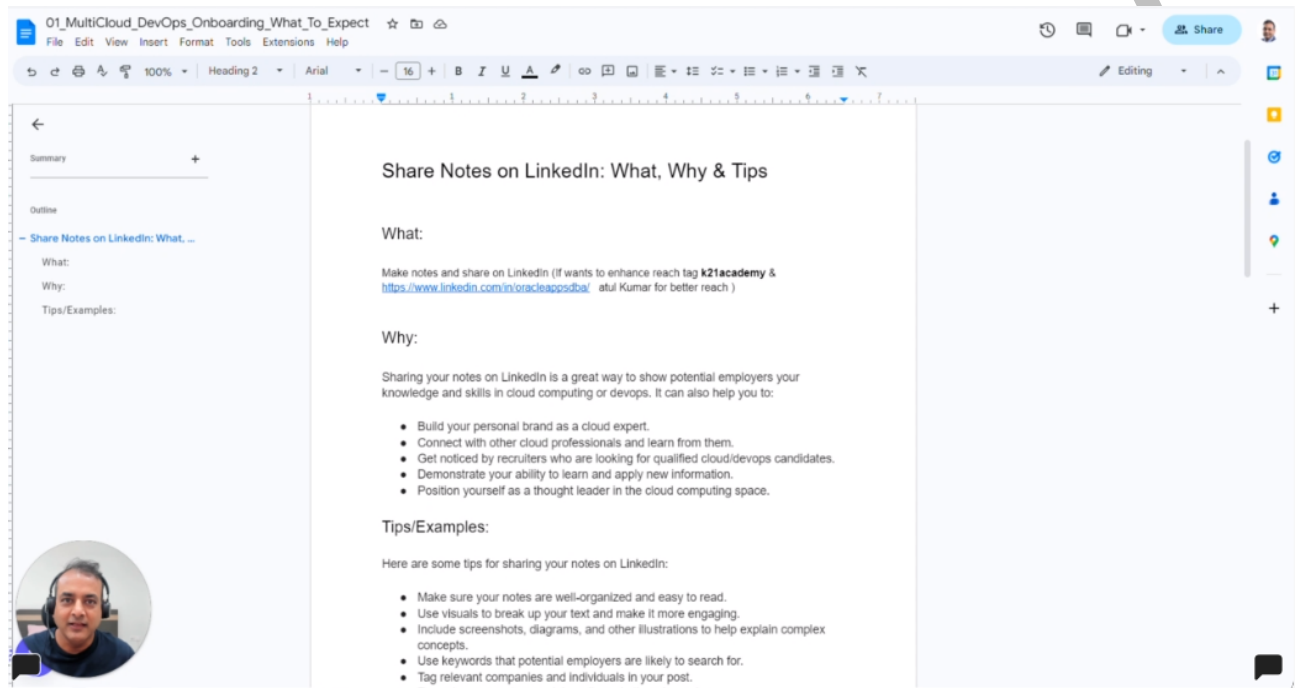
Note: If you Lost MFA Device or MFA Device Stops working then please refer to the **Troubleshooting Section**, where we have mentioned its fix.

6 SHARE YOUR LEARNINGS

In this section, you are going to share whatever you learned during this lab on LinkedIn and WhatsApp.

Please Watch the below 📺 video to understand how you can achieve this:

👤 <https://www.loom.com/share/d91f9abbec794f4f85e7d99bb82c2ff6>



👤 As you know, building a strong professional profile is essential in today's job market. One way to showcase your skills and knowledge is by sharing your labs and projects on LinkedIn.

Your LinkedIn profile is a powerful tool in your job search. Sharing your labs and projects is just one way to demonstrate your expertise and stand out to potential employers. If you don't have LinkedIn, we strongly recommend you create one for yourself.

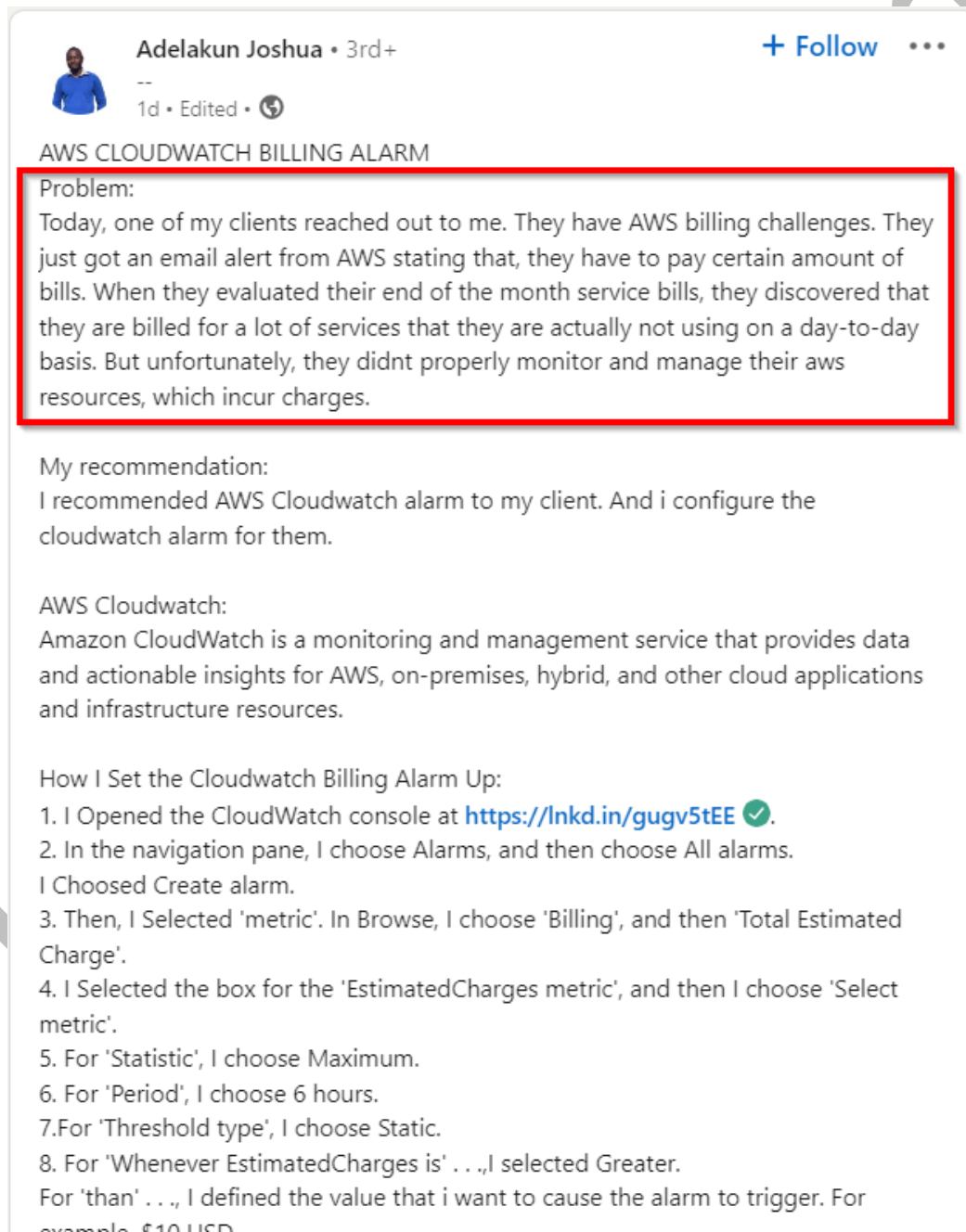
📝💻 By doing so, you can demonstrate to potential employers or connections that you have hands-on experience in your field and are actively engaged in learning and growing your skills.

6.1 On LinkedIn

Take a screenshot of the lab and share it on your LinkedIn. So, this will attract recruiters and employers to your profile and increase reach. **Do remember to tag K21Academy**, as we will circulate in our network too, to increase your reach.

Please refer the following examples on how to share on LinkedIn (don't limit yourself to just these types). Share learnings in points, notes 📝 or anything that can help others.

Here is a sample that includes steps inside a post followed by a screenshot:



SNS topic included the email address by which my client will receive email when the billing amount crosses the billing threshold specified.

Note:

You can select an existing Amazon SNS topic, create a 'new Amazon SNS topic', or use a topic ARN to notify other account. If you want your alarm to send multiple notifications for the same alarm state or for different alarm states, choose 'Add notification'.

11. I chose Next.

12. Under Name and description, I entered a name for the alarm. (Optional) Enter a description of the alarm.

13. Then I Chose 'Next'.

14. Under 'Preview and create', after confirming that the configuration is correct, and then I chose 'Create alarm'.

Conclusion: Organization that properly set up a cloudwatch billing alarm, will be able to properly monitor their aws resources usage. The metrics provided by the cloudwatch alarm through the dashboard will also help the organization to effectively control their spending.

#aws

#awscloud

#JTechconsult

#K21Academy: Learn Cloud From Experts

#AtulKumar

#SumtiMehta

Cloudwatch billing alarm architectural diagram:



Other examples:

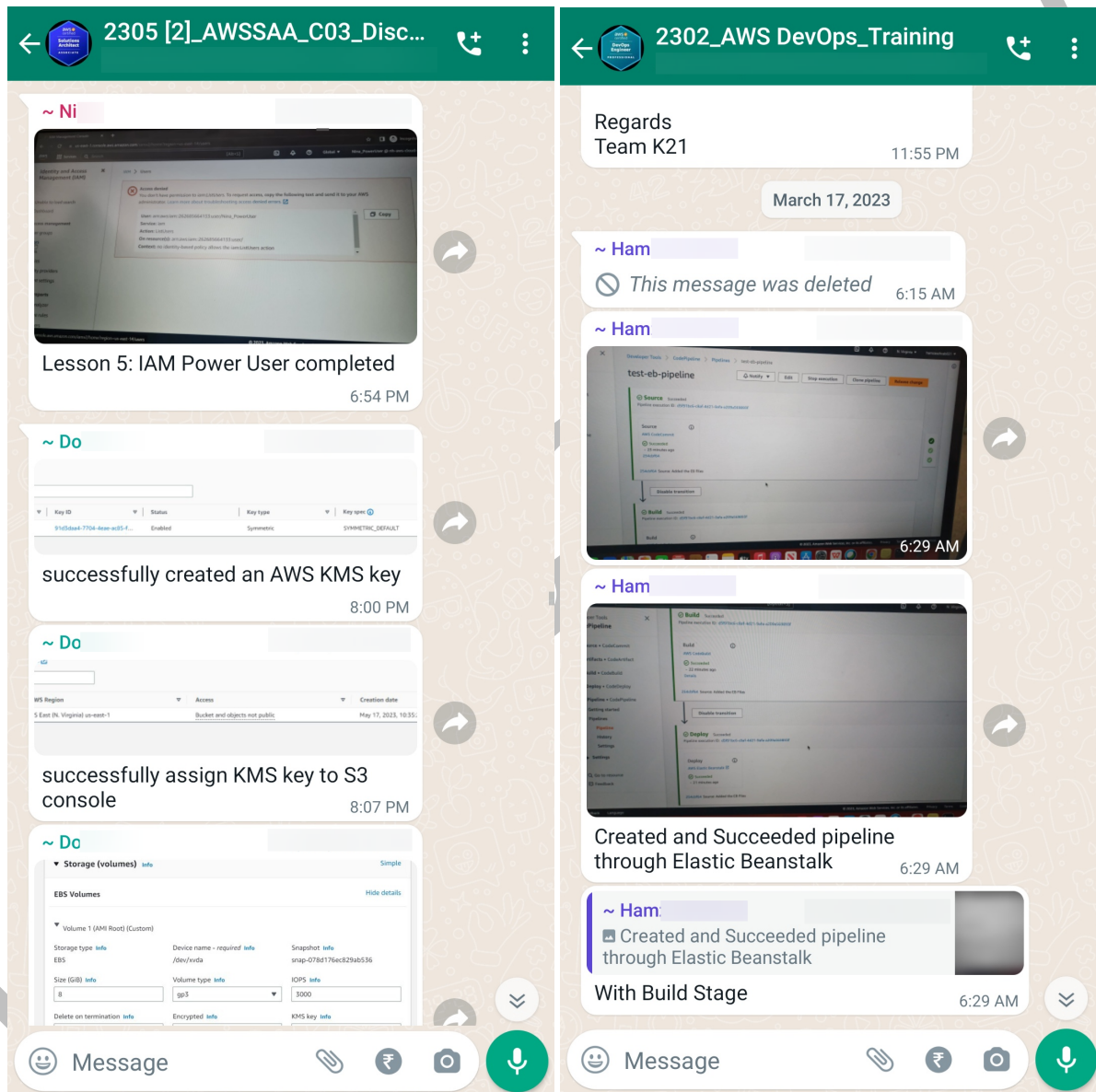
<https://bit.ly/3OvBC8s>

<https://bit.ly/47ZDwVL>

6.2 On WhatsApp

Share screenshot of lab in WhatsApp group so this will encourage others and also boost your confidence on the progressive path you are following.

Also, we will keep track of your progress on labs so do share your screenshot in WhatsApp group too.



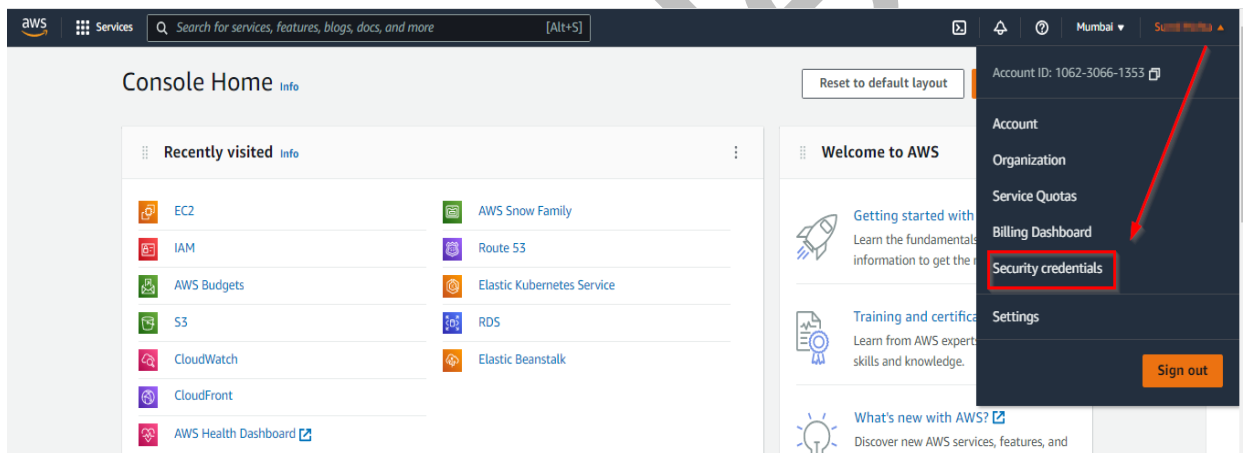
7 DELETING/CLEANUP

Note: In this section we will remove all the resources created by us as it might incurred the charges in your account.

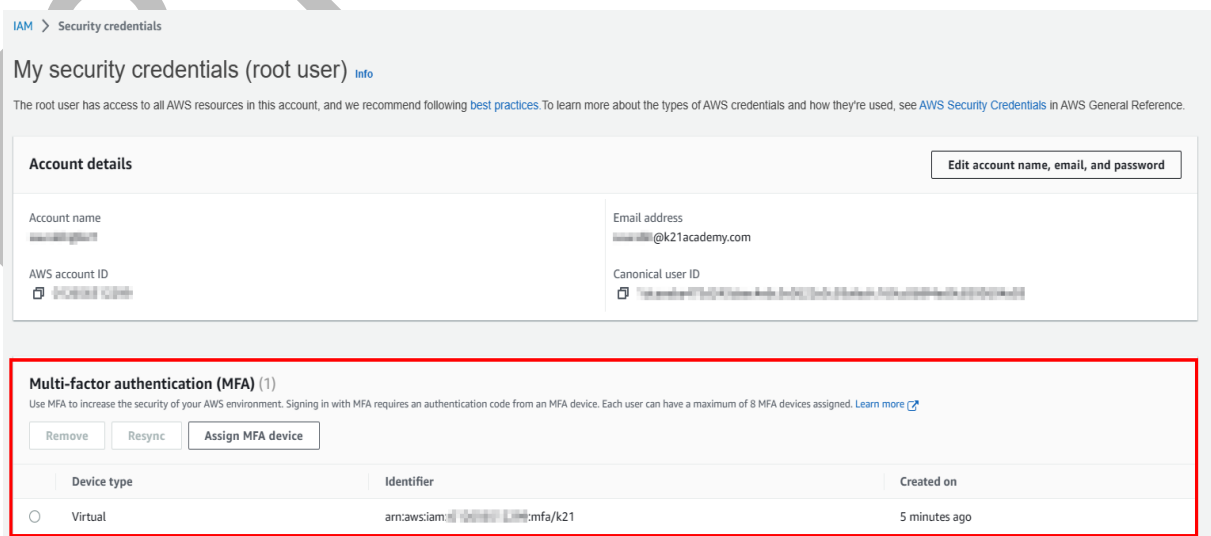
7.1 Deactivate the MFA Device

In this section, we will **deactivate** the Multi-Factor Authentication (MFA) from our AWS account. However, we **strongly recommend** that you follow the **security best practice** to enable multi-factor authentication (MFA) for your account. Because your root user can perform sensitive operations in your account, adding an **additional layer** of authentication helps you to better secure your account. If you still want to remove the MFA, please follow the below steps.

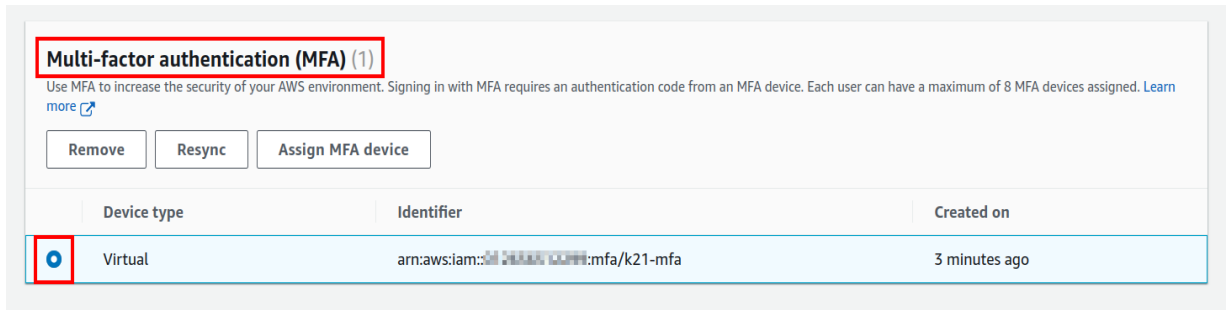
1. To deactivate the MFA device, click on **My Security Credential** tab.



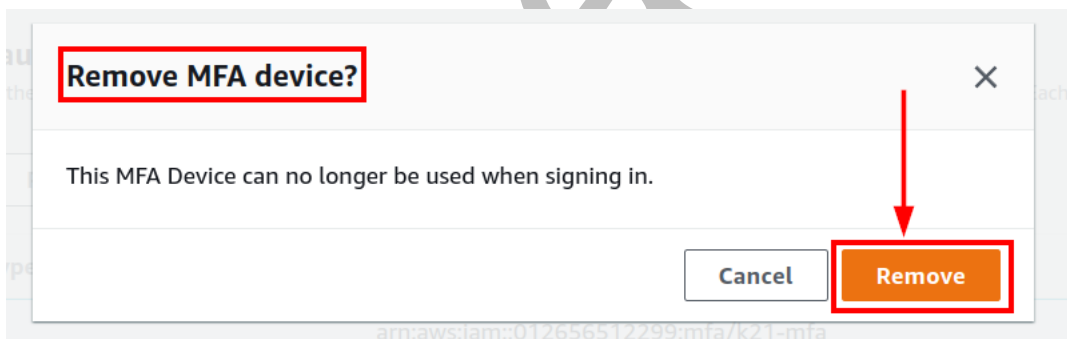
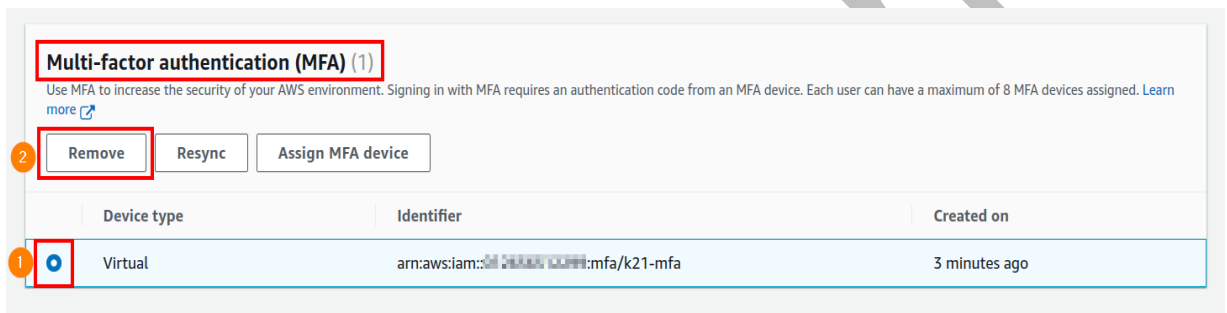
2. Now you shall see this screen.



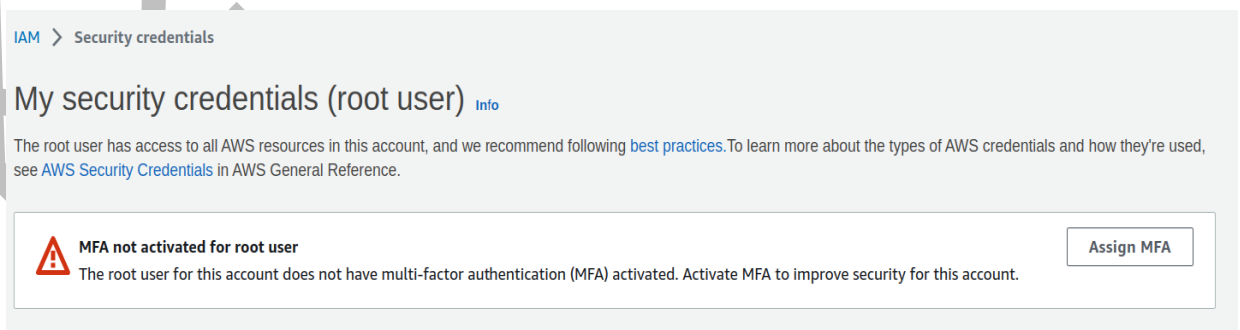
- Here the MFA is already activated, so we have to deactivate it. Click on the round circle to select our device first.



- Select **Remove** then a pop up will appear, click on **Remove**.



- Here we successfully **Deactivated** the MFA.



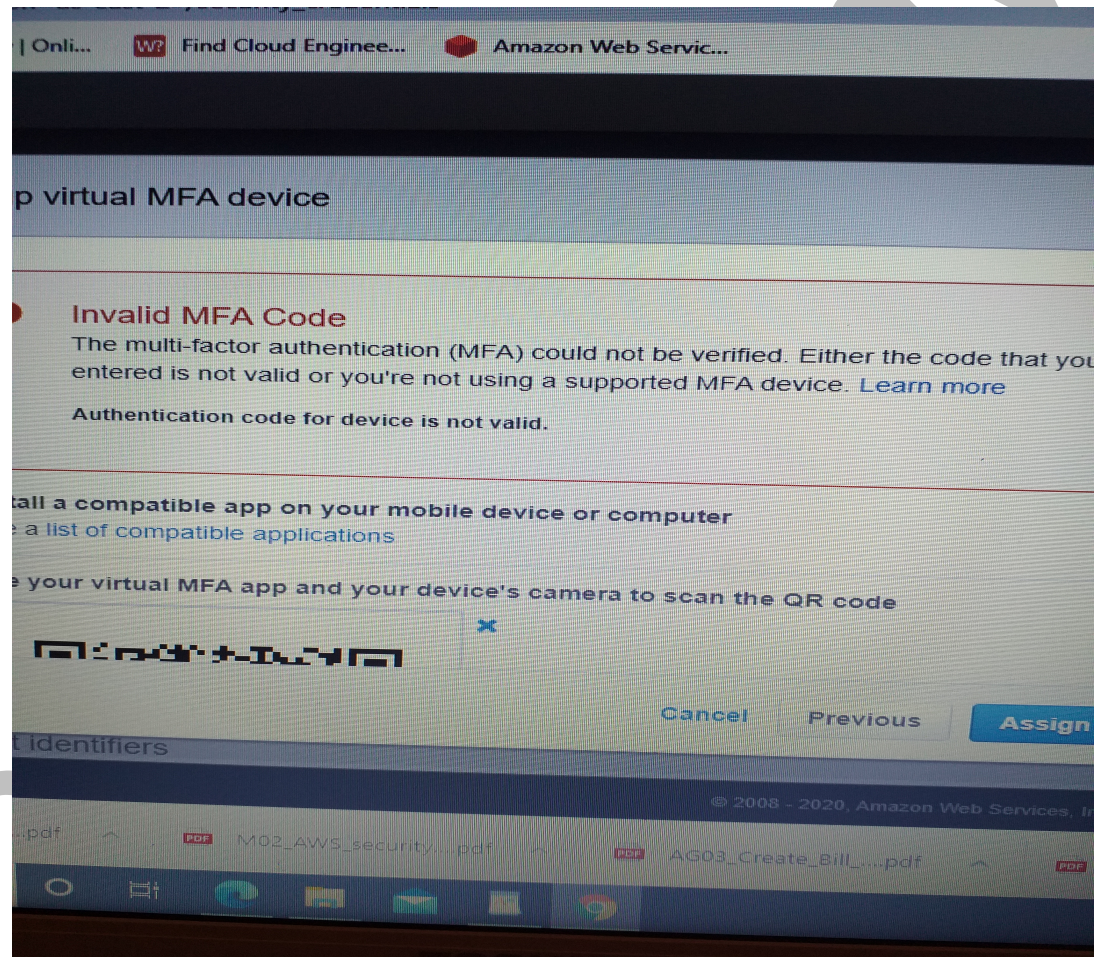
8 TROUBLESHOOTING

In this section we will be adding common error you might hit while performing the lab.

8.1 Invalid MFA Code Error

8.1.1 ERROR

Invalid MFA code error that shows the authentication code we enter is incorrect.



8.1.2 CAUSE

The link provided in the Activity guide is not working for you.

8.1.3 FIX

Try to install the Google Authenticator App directly from Google App Store rather than from the links provided in the guide, and the error shall be resolved.

8.2 Lost MFA Device or MFA Device Stops working

8.2.4 ERROR

Not able to Log into your AWS Account

8.2.5 CASUSE

You have lost your MFA Device or If your virtual MFA device or physical MFA device appears to be functioning properly, but you can't use it to access your AWS resources, it might be out of synchronization with AWS.

8.2.6 FIX

You can use alternative methods of authentication. This means that if you can't sign in with your MFA device, you can sign in by verifying your identity using the email and the primary contact phone number registered with your account.

Follow the below mentioned steps

1. Sign in to the AWS Management Console as the account owner by choosing Root user and entering your AWS account email address.
2. On the next page, enter your password. On the Additional verification required page, select an MFA method to authenticate with and choose Next.

3. On the **Multi-factor authentication** page, choose **Troubleshoot MFA**.



Multi-factor authentication

Your account is secured using multi-factor authentication (MFA). To finish signing in, turn on or view your MFA device and type the authentication code below.

Email address [REDACTED]

MFA code

Submit

Troubleshoot MFA

Cancel

4. On the **Troubleshoot your authentication device** page, in the **Sign in using alternative factors of authentication** section, choose **Sign in using alternative factors**.



Troubleshoot your authentication device

Re-sync with AWS servers

If your multi-factor authentication (MFA) device appears to be functioning properly, and you are not able to sign in, then the device might be out of sync.

Re-sync MFA device

Sign in using alternative factors of authentication

If your multi-factor authentication (MFA) device is lost, damaged, or not working, you can sign in using alternative factors of authentication. You must verify your identity using the email and phone registered with this account.

Sign in using alternative factors

[Sign in to a different account](#)

5. On the **Sign in using alternative factors of authentication** page, authenticate your account by verifying the email address, choose **Send verification email**.
6. Check the email that is associated with your AWS account for a message from Amazon Web Services (recover-mfa-no-reply@verify.signin.aws). Follow the directions in the email.
If you don't see the email in your account, check your spam folder, or return to your browser and choose **Resend the email**.
7. After you verify your email address, you can continue authenticating your account. To verify your primary contact phone number, choose **Call me now**.
8. Answer the call from AWS and, when prompted, enter the 6-digit number from the AWS website on your phone keypad.
9. After you verify your phone number, you can sign in to your account by choosing Sign in to the console.
10. The next step, remove the account from your device. Then go to the AWS Security Credentials page and delete the old MFA virtual device entity before you create a new one.

9 SUMMARY

This activity guide cover steps to:

- Enable MFA on Root account.
- Accessing AWS Console using MFA
- Deactivating of MFA